

# Yan Wang

Hongyi Honors Class, College of Life Sciences, Wuhan University

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## EDUCATION

### Wuhan University

September 2023-June 2027  
(expected)

B.S. in Biological Science | Hongyi Honors Class, College of Life Sciences

GPA: 3.91/4.00 · Major GPA: 3.97/4.00 · Rank: 1/33

Selected coursework: Cell Biology (95), Molecular Biology (96), Genetics (96), Biochemistry (95)

## RESEARCH EXPERIENCE

### Yale School of Medicine

July 2026-Present

Scherzer Lab, Stephen & Denise Adams Center for Parkinson's Disease Research

Visiting Undergraduate Researcher

#### PM20D1 and PM20D1-AS1 in Parkinson's disease

- Analyze PM20D1 and PM20D1-AS1 in public human brain RNA-seq datasets; assembled a PM20D1-AS1 tissue atlas and donor-by-cell-type expression summaries.
- Check transcript annotations and read support. Map qPCR primers and assess dilution series, amplification efficiency, and reverse-transcription controls before interpreting RNA measurements.
- Assist with iPSC culture and RNA experiments using SNCA-triplication and isogenic control lines.

### Wuhan University, Medical Research Institute

January 2025-Present

Joint Lab of Prof. Yanxun Yu and Prof. Younghan Jin

Undergraduate research

#### INS-1 signaling and ethanol avoidance in *C. elegans*

- Lead experiments on how INS-1 affects ethanol chemotaxis under different salt conditions, with guidance from my mentors.
- Use cell-specific gene deletion, rescue, neuronal ablation, and calcium imaging. Measured a shift in the ASER ethanol-response range and reduced AIY responses in ins-1 mutants.
- Compare insulin-pathway and ntr-1 mutants to investigate NTR-1 as a candidate receptor. Direct INS-1 binding and receptor activation remain untested.

#### Improving the POST-IT proximity-labeling system by directed evolution

- Construct PafA mutants and contribute to yeast display, protein purification, labeling assays, and flow cytometry.
- Helped screen an approximately  $10^6$ -variant library and test combined mutations at shorter labeling times and lower substrate concentrations.
- Contributed to testing conditions reduced from 18 h at 40  $\mu$ M substrate to 2 h at 10  $\mu$ M; use sequence and structural analysis to select mutations for follow-up.

### Wuhan University, College of Life Sciences

2024-2025

WHU-China iGEM Team

Project concept and prototype design; core team member

#### LOGIC - Lasting Optimized Genetic Input Calculator

- Proposed the concept and initial design for computation using bacterial colonies connected by diffusible quorum-sensing signals.
- Helped build and test response modules, logic gates, and a biological adder using spatially arranged sender and receiver colonies.
- Developed ImageJ macros for colony fluorescence analysis. The WHU-China team received an iGEM Gold Medal in 2025.

## MANUSCRIPTS & CONFERENCE ABSTRACTS

Wang, Y., Miller, L., Estrada, M., & Li, D.

**Mitochondrial dysfunction and  $\alpha$ -synuclein pathology in Parkinson's disease: A bidirectional pathogenic loop and therapeutic opportunities.**

*Frontiers in Molecular Neuroscience*. Review manuscript submitted.

Wang, Y., Zhang, J., Huang, X., Ruan, Y., & Yu, Y.

**Distributed INS-1 signaling through the candidate noncanonical receptor NTR-1 tunes the ASER-AIY axis to drive high-salt ethanol avoidance in *Caenorhabditis elegans*.**

*Neuroscience 2026, Society for Neuroscience*. Abstract submitted.

## SELECTED HONORS

SynBio Challenges - Gold Awards - Regular, Synthetic Cell, Protein Design, and Bacterial Game tracks 2026

Westlake University Rising Star Award 2026

iGEM Gold Medal - WHU-China team, Undergraduate Section 2025

Liu Daoyu Creative Learning Award, Second Prize - Wuhan University 2025

SynBio Challenges - Gold Award, Regular Track; Silver Award, Synthetic Cell Track 2025

Hongyi Honors College Annual Top Ten Scientific and Technological Advances Award - Wuhan University 2025

Outstanding Student Scholarship, Second Class - Wuhan University 2024, 2025

Outstanding Freshman Scholarship, First Class - Wuhan University 2023

## ACADEMIC ACTIVITIES

Peking University / Center for Life Sciences July 2026  
Summer Training Program in Neuroscience and Cognitive Science; Participant.

Westlake University July-August 2025  
Frontiers of Life Sciences International Undergraduate Summer School; Participant.

Tsinghua University May 2025  
National Forum on China's Top-Tier Talent Program 2.0 in Biology; Wuhan University student delegate.

## SKILLS

**Molecular and cell biology:** Molecular cloning, protein expression and purification, Western blotting, RT-qPCR, flow cytometry/FACS, mammalian cell culture, and iPSC handling.

**C. elegans experiments:** Strain construction, crosses, microinjection, chemotaxis assays, calcium imaging, optogenetic manipulation, and neuronal ablation.

**Data analysis:** R; bulk and single-nucleus RNA-seq analysis, transcript annotation and primer mapping, donor-level pseudobulk analysis, and scientific plotting.

**Structural analysis:** AlphaFold structure prediction, molecular docking, and PyMOL inspection.

**English:** TOEFL iBT: 106/120.